

Program: Diploma in Automation and Robotics	
Course Code:	Course Title: Python Programming
Semester: 3	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:3, T:0, P:0)	Periods per semester: 45

Course Objectives:

- To learn and understand Python programming basics and paradigm
- Discover the need for working with the strings and functions.
- Illustrate the process of structuring the data using lists, dictionaries, tuples and sets.
- Indicate the use of regular expressions and built-in functions to navigate the file system.
- To familiarize with the usage some commonly used python packages.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic principles and theorms of Engineering Mathematics		Mathematics I &II	1 & 2
Basic functions and features of Computer, Operating system and Internet applications, basic programming skills in Python.		Introduction to IT systems Lab	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand the broad concepts of robotics and its applications	10	Understanding
CO2	Understand and utilize the concept of decision making and loops to solve problems.	11	Applying

CO3	Apply functions, list, dictionary and string handling in Python to solve problems	12	Applying
CO4	Identify the commonly used operations involving file systems and regular expressions.	10	Applying
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	2					
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Make use of programming concepts with C programming.		
M1.01	Understand unique features of Python	1	Understanding
M1.02	Understand basic syntax of Python.	1	Understanding
M1.03	Use the concepts of Keywords, Variables, Constants, Data types.	3	Applying
M1.04	Identify output and input functions	2	Applying
M1.05	Identify different operators	3	Applying
Fundamentals of Python : History of Python - Unique features of Python - Basic Syntax - Variable and Data Types – Numbers – Strings – Lists – Tuples – Dictionary – Data type Conversion - Identifiers – Keywords – Indentation - Basic Operators - Arithmetic Operators - Comparison Operators - Assignment Operators - Bitwise Operators - Logical Operators - Membership Operators - Identity Operators – Boolean Operators – Statements –Expressions - Input and Output Operations			
CO2	Make use of iterative control structures and arrays in programs.		
M2.01	Identify the control structures – if, if-else nested if	4	Applying

M2.02	Identify loop structures – while, do-while, for and nested loops.	4	Applying
M2.03	Make use of unconditional control statements – break, continue, pass	3	Applying
	Series Test – I	1	
Python Program Flow - Decision Making - IF Statement - IF .ELIF ELSE Statement - Nested IF Statements - For loop using ranges, string, list and dictionaries - Use of while loops in python - Loop manipulation using pass, continue, break and else. – Sample Programs			
CO3	Develop programs using pointers and strings to solve problems more efficiently.		
M3.01	Understand functions, function call, arguments, return type	3	Applying
M3.02	Make use of recursive functions	1	Applying
M3.03	Familiarize with the concept of Strings and various string operations.	2	Applying
M3.04	Familiarize with the concept of lists and various list operations.	2	Applying
M3.05	Familiarize with the concept of Tuples and various tuple operations.	2	Applying
M3.06	Familiarize with the concept of Dictionary and various Dictionary operations.	2	Applying
Functions : Introduction, Function declaration and definition - Function call - Variable scope and lifetime - The return statement - Recursive functions – Strings – Concatenating, appending and multiplying strings – Immutability - String formatting operator – Built-in String Methods and function, Slice operation - Lists: access and update values in lists, nested and cloning lists, basic list operations, List methods, Using lists as Stack and Queues, list comprehensions. Tuple: Creating tuple, utility of tuples, accessing values in a tuple, updating tuple, deleting elements in tuple, basic tuple operations – Dictionary - Operations in Dictionary - Accessing Values in Dictionary - Updating Dictionary - Delete Dictionary Elements - Properties of Dictionary Keys - Built-in Dictionary Functions & Methods			
CO4	Make use of functions to solve problems effectively		
M4.01	Understand how to open and close data files in Python programs.	2	Understanding
M4.02	Understand how to Read and write data from/to files in Python programs.	2	Understanding
M4.03	Familiarize with the usage of different directory management methods.	2	Applying
M4.04	Familiarize with usage commonly used Python packages and libraries.	4	Applying

	Series Test – II	1	
File IO - File Operation - Opening and Closing Files - The open Function - The file Object Attributes - The close() Method - Reading and Writing Files - The write() and read() Method - File Positions - Renaming and Deleting Files rename(), The remove() Method - Directories in Python - mkdir(), chdir(), getcwd(), rmdir() - File & Directory Related Methods File Methods, close(), flush(),fileno(), isatty(),next(), read(), readline(), readlines(), seek(), tell(), truncate(), write(), writelines() – Modules - The import Statement - The from import Statement- Executing Modules as Scripts - Locating Modules - The PYTHONPATH Variable - Packages in Python – numpy – matplotlib etc			

Text / Reference

T/R	Book Title/Author
T1	ReemaThareja,“Python ProgrammingUsing Problem Solving Approach”, Oxford University Press, 2019.
R1	Core Python Programming, R. Nageswara Rao, Wiley India
R2	Python: The Complete Reference, Martin C. Brown, McGraw-Hill
R3	Introduction to Programming using Python, Y D. Liang, Pearson
R4	W.Chun , "Core Python Programming", 2nd Edition, Prentice Hall, 2006
R5	Jeeva Jose &P.SojanLal, “Introduction to Computing and Problem Solving with PYTHON”, Khanna Publishers, New Delhi, 2016

Online Resources

Sl.No	Website Link
1	Charles Severance: University of Michigan,Python for Everybody [COURSERA]. (05-01-2021), Available: https://www.coursera.org/
2	https://docs.python.org/3/tutorial/
3	https://www.w3schools.com/python/
4	Prof. SudarshanIyengar, IIT Ropar, Prof. Yayati Gupta, IIIT Dharwad, The Joy Of Computing Using Python [NPTEL], (05-01-2021), Available: https://nptel.ac.in/courses/106/106/106106182/#
5	Prof KannanMoudgalya, Professor, IIT Bombay, Python 3.4.3, [SWAYAM], (05-01-2021), Available: https://onlinecourses.swayam2.ac.in/aic20_sp33/preview